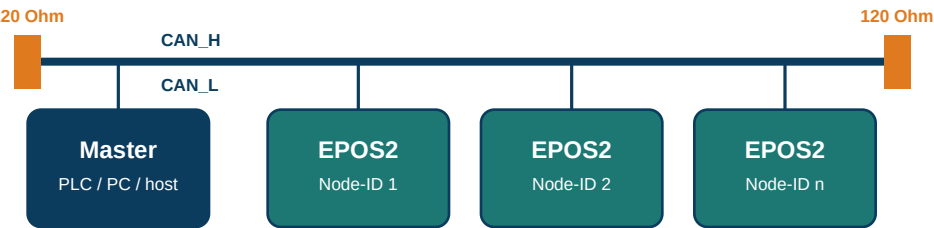


1 Bus and physical layer



Linear bus, short stubs. Terminate both ends only (EPOS2 has a DIP switch for on-board termination). Node-ID 1 to 127 (max 127 nodes). All nodes must use the same bit rate.

CAN bit rate (object 0x2001)

Code	Bit rate	Typ. max length
0	1 Mbit/s	25 m
1	800 kbit/s	50 m
2	500 kbit/s	100 m
3	250 kbit/s	250 m
4	125 kbit/s	500 m
5	50 kbit/s	1000 m
6	20 kbit/s	2500 m

Node-ID: DIP switch (1 to 15 on 24/2) or object 0x2000.
D-Sub 9: pin 7 CAN_H, pin 2 CAN_L, pin 5 shield.

2 Frame anatomy: the COB-ID is the whole addressing scheme

CAN data frame (standard 11-bit)

COB-ID (11 bit)	RTR	DLC	Data (0 to 8 bytes)
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Example: SDO request to Node-ID 3 = 0x600 + 3 = 0x603.

COB-ID = function code + Node-ID

Function code (4 bit)	Node-ID (7 bit)
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Low ID wins arbitration, so low function codes (NMT, SYNC, EMCY) outrank PDO, then SDO.

3 COB-ID map (default connection set)

Object	COB-ID	Direction	Purpose
NMT	0x000	master to node	Start / stop / reset nodes (CS byte + Node-ID, 0 = all)
SYNC	0x080	master to all	Triggers synchronous PDOs
EMCY	0x080 + ID	node to master	Emergency message with error code
TPDO 1	0x180 + ID	node to master	Default: Statusword (and position/velocity, if mapped)
RPDO 1	0x200 + ID	master to node	Default: Controlword (and target values, if mapped)
TPDO 2 / RPDO 2	0x280 / 0x300 + ID	both	Freely mappable process data
TPDO 3 / RPDO 3	0x380 / 0x400 + ID	both	Freely mappable process data
TPDO 4 / RPDO 4	0x480 / 0x500 + ID	both	Check your model: PDO count varies by firmware
SDO (node to master)	0x580 + ID	node to master	Response to an SDO request
SDO (master to node)	0x600 + ID	master to node	Read / write any object dictionary entry
Heartbeat / Boot-up	0x700 + ID	node to master	NMT state broadcast, boot-up message

priority: high to low

4 Why this matters

Two ways to talk

SDO for configuration and one-off reads (slow, confirmed). PDO for cyclic control (fast, no overhead).

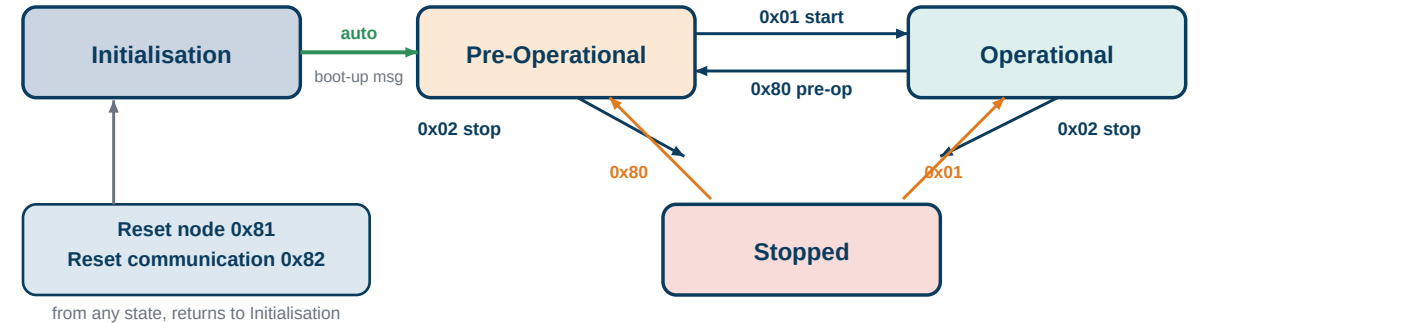
PDOs are gated by NMT

PDO traffic only flows when the node is Operational. SDO works in Pre-Operational and Operational.

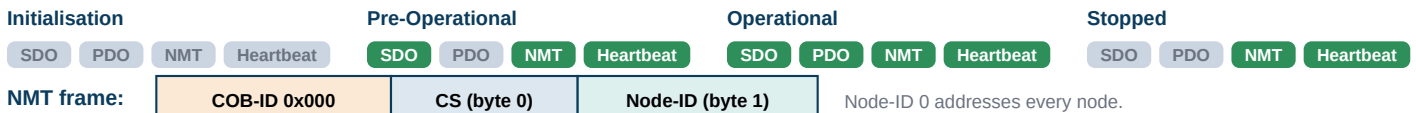
Everything is an object

Every parameter lives at index/sub-index in the object dictionary. The next pages show how to reach them.

1 NMT state machine



What is allowed in each state



Heartbeat (0x700 + ID), data byte: 0x00 boot-up, 0x04 stopped, 0x05 operational, 0x7F pre-operational

2 SDO: read or write any object (expedited transfer)

Request: COB-ID 0x600 + Node-ID

CS	Index LSB	Index MSB	Sub	Data 0	Data 1	Data 2	Data 3	All values little endian.
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Response: COB-ID 0x580 + Node-ID

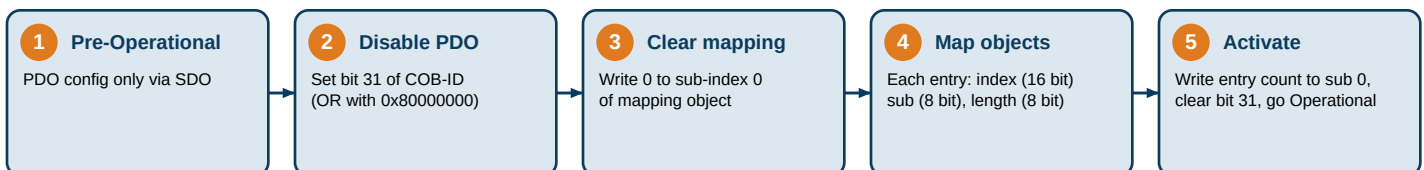
CS	Index LSB	Index MSB	Sub	Data 0	Data 1	Data 2	Data 3	Abort: CS 0x80, data = abort code.
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Command	CS	Meaning	Abort code	Meaning
Write 1 / 2 / 3 / 4 byte	0x2F / 0x2B / 0x27 / 0x23	Download request	0x06020000	Object does not exist
Write OK	0x60	Confirmation	0x06010002	Write to read-only object
Read request	0x40	Upload request	0x06040041	Object not mappable to PDO
Read reply 1 / 2 / 3 / 4 byte	0x4F / 0x4B / 0x47 / 0x43	Data bytes 4 to 7	0x0F00FFB9	Wrong CAN ID (maxon)
			0x0F00FFC0	Wrong NMT state (maxon)

Example, node 1: write Controlword (0x6040) = 0x000F

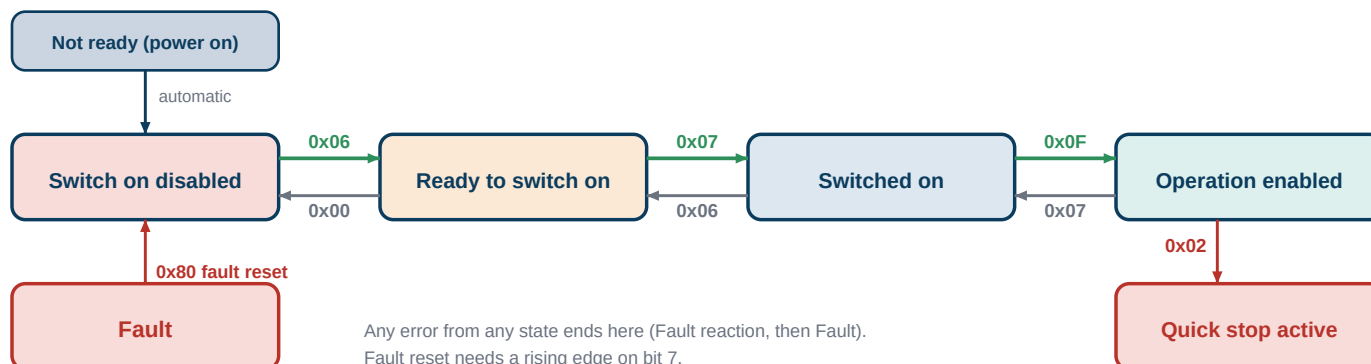
ID 0x601 data 2B 40 60 00 0F 00 00 00 reply: ID 0x581 data 60 40 60 00 00 00 00 00

3 PDO: fast cyclic process data



Object	Index	Transmission type	Behaviour
RPDO comm. param.	0x1400 to 0x1403	1 to 240	Every n-th SYNC
RPDO mapping	0x1600 to 0x1603	254 / 255	Asynchronous, on event / change
TPDO comm. param.	0x1800 to 0x1803	RTR	Sent only on remote request
TPDO mapping	0x1A00 to 0x1A03		

1 Device state machine (Controlword 0x6040 drives it)



Statusword 0x6041 (what the drive reports)

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Pos. ref. found		Mode specific	Mode specific	Limit	Target reached	Remote		Warning	Switch on disabled	Quick stop	Voltage enabled	FAULT	Op enabled	Switched on	Ready

2 Operating modes and key objects

Value	Mode (object 0x6060)
1	Profile Position (PPM)
3	Profile Velocity (PVM)
6	Homing
7	Interpolated Position
-1 (0xFF)	Position (PM)
-2 (0xFE)	Velocity (VM)
-3 (0xFD)	Current (CM)
-4 / -5	Master encoder / Step-Dir

Index	Object
0x1017	Producer heartbeat time
0x2000	Node-ID
0x2001	CAN bit rate
0x6040 / 0x6041	Controlword / Statusword
0x6060 / 0x6061	Mode set / mode display
0x607A / 0x6064	Target / actual position
0x60FF / 0x606C	Target / actual velocity
0x6081 / 0x6083 / 0x6084	Profile velocity / accel / decel

3 Bring-up sequence for node 1: move to position 0x1000 (PPM)

#	Action	COB-ID	Data (hex)
1	NMT start node 1 (Operational)	0x000	01 01
2	Mode of operation = Profile Position	0x601	2F 60 60 00 01
3	Controlword: Shutdown	0x601	2B 40 60 00 06 00
4	Controlword: Switch on	0x601	2B 40 60 00 07 00
5	Controlword: Enable operation	0x601	2B 40 60 00 0F 00
6	Target position 0x607A = 0x00001000	0x601	23 7A 60 00 00 10 00 00
7	Controlword 0x3F: start, absolute, immediate	0x601	2B 40 60 00 3F 00
8	Watch Statusword bit 10 (target reached)	0x581	read 0x6041 (CS 0x40)

Tips Each SDO write is acknowledged (0x581 reply with CS 0x60). Wait for it before the next frame.
For continuous control, map Controlword and target values to an RPDO and Statusword and position to a TPDO.
Set the mode (step 2) before enabling the drive. Clear faults with 0x80 before retrying.